

Chapter 10.1

Potential Vegetation and Floristic Composition of Andean Forests in South Ecuador, with a Focus on the RBSF

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10.1.1 Introduction: Andean Forests

A decrease in overall plant species diversity and forest stature with increasing elevation is a general trend in moist tropical mountain areas (e.g., Aiba and Kitayama 1999; Cavelier 1996; Gentry 1988, 1995; Grubb 1977; Lieberman et al. 1996; Richter 2001) and was confirmed for the RBSF (Homeier 2004).

However, the mechanisms responsible for the altitudinal change in both plant diversity and forest structure in tropical mountain forests remain poorly understood (see Chapters 10.3, 10.4 in this volume).

The altitudinal gradient, as well as the topographical gradient at a smaller scale, combines sets of environmental factors such as soil hydrology, nutrient supply, temperature and wind exposure, which are difficult to disentangle. The environmental abiotic settings of the study area are described by Bendix et al. (Chapter 8), Richter and Moreira-Muñoz (2005) and Wilcke et al. (Chapter 9).

Ecuadorian woody plant species richness declines with increasing elevation from the Amazonian lowlands, which harbour the world's highest alpha diversity in trees (Balslev et al. 1998), to upper montane forests (Jørgensen and León-Yáñez 1999). Comparing Neotropical forests, Gentry (1988) found no pronounced differences in woody plant diversity for sites below ca. 1500 m, but a general decrease above. In contrast, epiphyte species richness peaks in montane forests: around 1000–2500 m a.s.l. in vascular epiphytes (Ibisch et al. 1996; Küper et al. 2004; Krömer et al. 2005) and between 2500–3000 m a.s.l. in bryophytes (Wolf 1993). Family patterns of species richness are fairly contrasting, but many are hump-shaped with maxima at middle elevations (Kessler 2002a). The same is true for elevational patterns of endemism (Kessler 2002a). In general, highest percentages of endemism for the Ecuadorian vascular plant flora are found in the Andean forests above 1500 m a.s.l. (Jørgensen and León-Yáñez 1999; Kessler 2002a). The Podocarpus National Park in particular is known for its high endemism with more than 200 species restricted to this area (Valencia et al. 2000).

10.1.2 Results and Discussion

10.1.2.1 *Potential Vegetation of the Podocarpus National Park Area*

A classification of the plant formations was compiled using information from former publications, maps and our own records (Aguirre et al. 2003; Diels 1937; Grubb et al. 1963; Grubb and Whitmore 1966; Kessler 1991; Madsen and Øllgaard 1993; Ulloa and Jørgensen 1995; Valencia et al. 1999; Balslev and Øllgaard 2002; Lozano 2002; Richter 2003). Seven major types of potential natural vegetation are distinguished for the wider study area (see Fig. 1.6 in Chapter 1 in this volume). We omit the description of a variety of additional main vegetation types in south-western Ecuador (west of easternmost Loja province and the upper Rio Catamayo drainage) such as coastal dry forests or cloud forests, being not of major relevance for our study site in terms of floristic interchange. There is of course a great variety of vegetation subtypes, since not only the floristic composition or climatic gradients drive the differentiation of communities, but also edaphic factors, diverging traits of vegetation history and multiple disturbance regimes. Not all of these subtypes can be treated in the context of this short overview.

Currently, much of the province of Zamora–Chinchipec and most of the neighbouring province of Loja is stripped of its natural vegetation. For many centuries, but especially in the course of the past decades, large parts of the study area were converted. The main current threats for the forests of the region are extraction of timber, escaping fires and especially conversion into cattle pasture. With the construction of further roads, deforestation expands and the last remote territories are opened up to colonization. Quite likely, only a few patches within protected areas such as the Podocarpus National Park will survive the coming decades at the current rate of conversion. The vegetation types are specified in Table 10.1.1.

10.1.2.2 *Floristic Composition of the RBSF*

To date, 1208 species in 425 genera and 131 families of native spermatophytes have been recorded from the RBSF (Homeier and Werner 2007). This list remains fragmentary; sampling is incomplete, especially as non-woody plants and higher altitudes have only been collected opportunistically. We estimate the 1000 ha reserve harbours some 1500–1700 species of seed plants, or about 10% of the entire spermatophyte flora of Ecuador.

Both in terms of physiognomy and floristic composition, two steep local gradients promote the coexistence of a wide array of vegetation types and microhabitats. With increasing altitude, height and closure of the canopy quickly decline, coupled with high species turnover. At an even smaller spatial scale, corresponding changes take place along the ravine–slope–ridge gradient (Homeier 2004; see Chapter 18 in this volume).

Table 10.1.1 The seven vegetation types depicted in Fig. 1.1.6 and some of their characteristic vascular plant species (altitudinal ranges of the types strongly depend on local conditions)

- A Evergreen premontane (= submontane) rainforest around Zamora, ca. 800–1300 m a.s.l.; tierra subtemplada, 12 humid months.
This forest can attain up to 40 m height. The few characteristic species include: *Ceroxylon amazonicum*, *Euterpe precatoria*, *Iriartea deltoidea* and *Oenocarpus bataua* (Arecaceae), *Ochroma pyramidale* (Bombacaceae), *Terminalia amazonia* and *T. oblonga* (Combretaceae), *Grias peruviana* (Lecythidaceae), *Clarisia racemosa*, *Ficus* spp and *Sorocea trophoides* (Moraceae), *Otoba glycarpa* (Myristicaceae), *Piper* spp (Piperaceae), *Palicourea guianensis*, *Psychotria* spp (Rubiaceae), *Pouteria* sp. (Sapotaceae).
- B Evergreen lower montane forest ca. 1300–2100 m a.s.l. on the eastern escarpment of the Cordillera Real; tierra subtemplada and templada, 12 humid months.
Canopy height up to 30 m. Characteristic species: *Alzatea verticillata* (Alzateaceae), *Chamaedorea pinnatifrons*, *Dictyocaryum lamarckianum* and *Wettinia maynensis* (Arecaceae), *Piptocoma discolor* and *Mikania* spp (Asteraceae), *Tabebuia chrysantha* (Bignoniaceae), *Vismia tomentosa* (Clusiaceae), *Cyathea caracasana* (Cyatheaceae), *Inga acreana* and other *Inga* spp (Fabaceae), *Nectandra lineatifolia*, *N. membranacea* and *Ocotea aciphylla* (Lauraceae), *Miconia imitans*, *M. punctata* (Melastomataceae), *Cedrela montana* (Meliaceae), *Ficus* spp, *Morus insignis* and *Sorocea trophoides* (Moraceae), *Piper* spp (Piperaceae), *Heliocarpus americanus* (Tiliaceae).
- C Evergreen upper montane forest on the eastern escarpment above Sabanilla, ca. 2100–2700 m a.s.l.; tierra fresca, 12 humid months.
Canopy heights attain up to 25 m. Characteristic species: *Ilex rimbachii* (Aquifoliaceae), *Hedyosmum* spp (Chloranthaceae), *Clethra revoluta* (Clethraceae), *Clusia ducu* and *Tovomita weddeliana* (Clusiaceae), *Weinmannia pinnata* and other *Weinmannia* spp (Cunoniaceae), *Cyathea bipinnatifida* (Cyatheaceae), *Purdiaea nutans* (Clethraceae), *Bejaria aestuans* (Ericaceae), *Alchornea grandiflora* (Euphorbiaceae), *Macrocarpaea revoluta* (Gentianaceae), *Eschweilera sessilis* (Lecythidaceae), *Licaria subsessilis*, *Ocotea benthamiana* and *Persea ferruginea* (Lauraceae), *Graffenrieda emarginata*, *G. harlingii* and *Tibouchina lepidota* (Melastomataceae), *Myrica pubescens* (Myricaceae), *Myrsine coriacea* (Myrsinaceae), *Calyptranthes pulchella* and *Myrcia* spp (Myrtaceae), *Podocarpus oleifolius* and *Prumnopitys montana* (Podocarpaceae), *Dioicodendron dioicum* and *Palicourea* spp (Rubiaceae), *Matayba inelegans* (Sapindaceae), *Drimys granadensis* (Winteraceae).
- D Evergreen elfin-forest (ceja de montaña) on the eastern and western escarpment of the Cordillera Real up to the timberline, ca. 2700–3100 m a.s.l.; tierra fría, 12 humid months.
This forest type forms the timberline. Canopy height up to 6–8 m, rarely higher. Characteristic species: *Ilex* spp (Aquifoliaceae), *Puya eryngioides* (Bromeliaceae), *Hedyosmum cumbalense* and *H. scabrum* (Chloranthaceae), *Clethra ovalifolia* (Clethraceae), *Clusia elliptica* (Clusiaceae), *Weinmannia cochensis*, *W. loxensis* and *W. rollottii* (Cunoniaceae), *Gaultheria reticulata* (Ericaceae), *Escallonia myrtilloides* (Grossulariaceae), *Orthrosanthus chimborazensis* (Iridaceae), *Persea ferruginea* and *Ocotea infraveolata* (Lauraceae), *Gaiadendron punctatum* (Loranthaceae), *Graffenrieda harlingii* (Melastomataceae), *Myrteola phyllicoides* (Myrtaceae), *Hesperomeles ferruginea* (Rosaceae), *Styrax foveolaria* (Styracaceae), *Symplocos sulcinervia* (Symplocaceae), *Gordonia fruticosa* (Theaceae), as well as impenetrable bamboo of *Chusquea falcata* and *C. scandens* (Poaceae).
- E Shrub and dwarf bamboo páramos in the crest region of the Cordillera Real above timberline, ca. 3100–3700 m a.s.l.; tierra subhelada, 12 humid months.
Also referred to as jalca in a physiognomical sense. This vegetation attains heights of up to 2 m. Characteristic species: *Gynoxis* spp (Asteraceae), *Puya eryngioides* and *Puya nitida* (Bromeliaceae), *Hypericum decandrum* (Clusiaceae), *Rhynchospora vulcani* (Cyperaceae), *Bejaria resinosa*, *Disterigma pentandrum*, *Gaultheria erecta*, *G. reticulata* and *Vaccinium floribundum* (Ericaceae), *Escallonia myrtilloides* (Grossulariaceae), *Brachyotum andreaeanum* (Melastomataceae), *Neurolepis asymmetrica*, *N. elata*, *N. laegaardii* and the bamboo species

(continued)

Table 10.1.1 (continued)

- Chusquea neurophylla* (Poaceae), *Monnina arbuscula* (Polygalaceae), *Valeriana microphylla* and *V. plantaginea* (Valerianaceae).
- F Semideciduous interandean forest in the valley region west of the Cordillera Real around Malacatos, Vilcabamba and Yangana, ca. 1400–2400 m a.s.l.; tierra templada and fresca, 6–8 humid months.
Canopy height up to 15 m, emergents occasionally up to 20 m. Characteristic species: *Cacosmia rugosa*, *Verbesina lloensis* and *Zexmenia helianthoides* (Asteraceae), *Tabebuia chrysantha* (Bignoniaceae), *Ceiba insignis* (Bombacaceae), *Cordia macrocephala* (Boraginaceae), *Buddleja lojensis* (Buddlejaceae), *Capparis scabrida* (Capparidaceae), *Croton wagneri* (Euphorbiaceae), *Acacia macracantha*, *Anadenanthera colubrina* (Fabaceae), *Lepechinia mutica* and *Scutellaria volubilis* (Lamiaceae), *Adenaria floribunda* (Lythraceae), *Cantua quercifolia* (Polemoniaceae), *Coccoloba ruiziana* (Polygonaceae), *Oreocallis grandiflora* and *Roupala obovata* (Proteaceae), *Dodonaea viscosa* (Sapindaceae), *Streptosolen jamesonii* (Solanaceae).
- G Evergreen upper montane forest on the western escarpment above Loja and Yangana, ca. 2400–2800 m a.s.l.; tierra templada and fresca, 8–11 humid months.
Canopy height up to 20 m. Characteristic species: *Hedyosmum racemosum* (Chloranthaceae), *Clusia alata*, *C. elliptica* (Clusiaceae), *Weinmannia glabra*, *W. macrophylla* (Cunoniaceae), *Persea ferruginea* (Lauraceae), *Graffenrieda harlingii* (Melastomataceae), *Cedrela montana*, *Ruarea hirsuta* (Meliaceae), *Myrica pubescens* (Myricaceae), *Ardisia* sp., *Myrsine andina* (Myrsinaceae), *Myrcianthes rhopaloides* (Myrtaceae), *Prumnopitys montana* (Podocarpaceae), *Oreocallis grandiflora* (Proteaceae), *Cinchona macrocalyx* and *C. officinalis*, *Palicourea heterochroma* (Rubiaceae).

The diversity of vegetation types is further increased by tracts of secondary vegetation (mostly at 1800–1950 m a.s.l.) in different stages of recovery from clear-cutting and the construction of a pipe system for the close-by hydroelectric plant, respectively.

Finally, the study area constitutes a well developed mosaic of natural successional stages due to tree fall gaps (see Chapter 23) and landslides, which are conspicuously common throughout the whole area (see Chapter 24). However, successional habitats of natural or anthropogenic origin are not treated in this chapter. The timberline is poorly defined and highly variable in altitude (see Chapter 10.4); the forest decreases in stature and opens gradually giving way to shrub páramo at ca. 2600–2700 m a.s.l. along exposed ridges and considerably higher in major ravines (ca. 3000 m a.s.l.).

The most important seed plant families are Orchidaceae (337 spp), Rubiaceae (65 spp), Bromeliaceae (64 spp) and Melastomataceae (55 spp; Homeier and Werner 2007). So far, more than 280 tree species (diameter at breast height exceeding 5 cm) are known from the RBSF; the most speciose families are Lauraceae, Melastomataceae and Rubiaceae. *Graffenrieda emarginata* (Melastomataceae; see Homeier 2005a), *Purdiaea nutans* (Clethraceae; see Homeier 2005b), *Alchornea grandiflora* (Euphorbiaceae) and *Podocarpus oleifolius* (Podocarpaceae) are some of the most common and characteristic tree species of the RBSF forest.

About 140 climber species are known to occur in the RBSF, the most important families being Asteraceae, Apocynaceae *s.l.* and Araceae.

More than 400 angiosperm epiphytes (the largest families being Orchidaceae, Bromeliaceae, Piperaceae) have been registered. Werner et al. (2005) found up to 98 species of vascular epiphytes on single trees – one of the highest numbers recorded worldwide (see Schuettelpelz and Trapnell 2006).

Fern and fern allies with 257 species (Lehnert et al. 2007) and bryophytes with more than 515 species (Gradstein et al. 2007; Parolly et al. 2004) are equally speciose. The number of bryophyte species is the highest recorded from such a small tropical area and constitutes about one-third of the entire bryophyte flora of Ecuador (Léon et al. 2006).

Four main forest types were distinguished for the lower reaches of RBSF by their tree species composition (Fig. 10.1.1; Homeier 2004; Homeier et al. 2002). These were confirmed by studies on the syntaxonomy of epiphytic bryophytes (Parolly and Kürschner 2004b) and on climber species composition (Matezki, in preparation). The less accessible vegetation types such as ravine forest above 2200 m a.s.l. remain to be studied in detail.

The following description lists some structural features and characteristic plant species for each of the four forest types and for the shrub páramo that occurs at the highest elevations of the RBSF.

10.1.2.2.1 Forest of the San Francisco Valley and Major Ravines Below ca. 2200 m a.s.l. (Forest Type I, Subtype of B. Evergreen Lower Montane Forest)

The tallest and most speciose forest grows on gentle lower slopes of the San Francisco river and within major ravines where the canopy attains 20–25 m with emergents occasionally approaching heights of 35 m (Fig. 10.1.2). Megaphyllous shrubs (e.g., Piperaceae) and herbs (Cyclanthaceae, Heliconiaceae, Zingiberaceae) abound in the understorey. Terrestrial and bole climbing Araceae are both common and speciose. Understorey epiphytes are predominantly skiophilous ferns, especially Aspleniaceae (*Asplenium auritum*, *A. theciferum*), Dryopteridaceae (*Elaphoglossum* spp, *Polybotria* spp), Hymenophyllaceae (*Trichomanes angustatum*, *T. diaphanum*, *T. reptans*) and Vittariaceae (*Polytaenia brasilianum*, *P. lineatum*) as well as *Guzmania confusa* and *G. morreniana* (Bromeliaceae). Due to low light levels generalistic and heliophilous epiphytes (e.g., Ericaceae, Tillandsia spp, Orchidaceae) are virtually confined to higher strata.

Several important families of moisture loving herbs show a relatively low diversity and abundance at RBSF and are largely restricted to the understorey of this forest type (e.g., Begoniaceae, Campanulaceae, Gesneriaceae).

Characteristic trees include *Piptocoma discolor* (Asteraceae), *Tabebuia chrysantha* (Bignoniaceae), *Hyeronima asperifolia* and *Sapium glandulosum* (Euphorbiaceae), *Nectandra linneatifolia* and *N. membranacea* (Lauraceae), *Meriania* sp., *Miconia punctata* and other *Miconia* spp (Melastomataceae), *Inga*

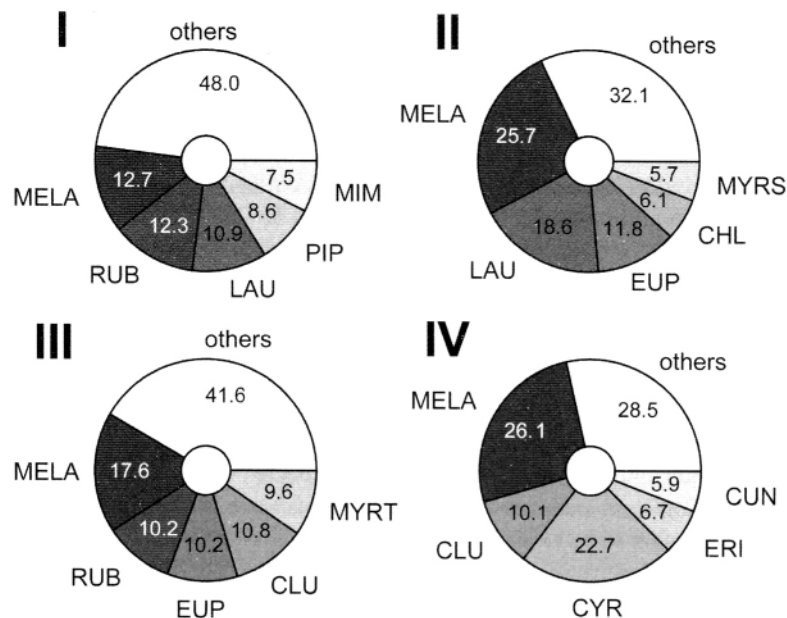


Fig. 10.1.1 Floristic composition of trees (dbh ≥ 5 cm) throughout the four forest types (I–IV) from the RBSF described in detail. Numbers indicate the respective percentages of total tree species richness (after Homeier 2004). *CHL* Chloranthaceae, *CUN* Cunoniaceae, *CLU* Clusiaceae, *CYR* Cyrillaceae, *ERI* Ericaceae, *EUP* Euphorbiaceae, *LAU* Lauraceae, *MELA* Melastomataceae, *MIM* Mimosaceae, *MYRS* Myrsinaceae, *MYRT* Myrtaceae, *PIP* Piperaceae, *RUB* Rubiaceae

spp (Mimosaceae), *Morus insignis*, *Naucleopsis* sp. nov., and *Ficus* spp (Moraceae), *Prumnopitys montana* (Podocarpaceae), *Micropholis guyanensis* (Sapotaceae) and *Heliocarpus americanus* (Tiliaceae).

Predominating shrubs include *Saurauia* spp (Actinidiaceae), *Chamaedorea pinnatifrons* (Arecaceae), *Piper* spp (Piperaceae), *Farama* spp, *Palicourea* spp and *Psychotria* spp (notably *P. tinctoria*; Rubiaceae).

Bryophytes are very abundant and speciose in this forest type and cover tree trunks and branches with thick mats. Characteristic epiphytes in the forest understorey and lower canopy include numerous species of the liverwort genus *Plagiochila* (e.g., *P. bryopteroides*, *P. deflexirama*, *P. diversifolia*, *P. heterophylla*, *P. montagnei*, *P. oresitropha*, *P. punctata*, *P. stricta*, *P. superba*), furthermore the feather-like *Bryopteris filicina* and *Porella crispata*, the pendent *Radula voluta* and *Taxilejeunea pterigonia* and robust mosses such as *Porotrichum expansum*, *Porotrichodendron superbum*, *Prionodon densus*, *Pterobryon densum* and various pendent Meteoriaceae. Tree bases harbour large mats of *Trichocolea tomentosa* and *Rhizogonium spiniforme* and tree canopies are rich in species of *Frullania* (e.g., *F. brasiliensis*, *F. intumescens*, *F. kunzei*, *F. pittieri*, *F. arecae*). Thalloid liverworts of *Symphyogyna* (*S. aspera*, *S. brasiliensis*, *S. brogniartii*) and *Riccardia andina* are characteristic on forest soil.



Fig. 10.1.2 Profile diagram of forest type I (Forest of the San Francisco valley and major ravines below ca. 2200 m a.s.l., subtype of B. Evergreen lower montane forest) at the RBSF (1960 m a. s.l., depth of the profile is 5 m). Trees with diameters exceeding 5 cm: 1 *Talauma* sp., 2 *Ficus citrifolia*, 3 *Morus insignis*, 4 *Sapium glandulosum*, 5 Lauraceae, 6 *Inga* sp. 1, 7 *Prumnopitys montana*, 8 Myrsinaceae, 9 *Miconia theaezans*, 10 *Ilex* cf. *amboroica*, 11 *Ocotea* sp., 12 Lauraceae, 13 *Micropholis guyanensis*, 14 *Miconia quadriporata*, 15 *Inga* sp. 2, 16 *Ficus tonduzii*, 17 *Psychotria tinctoria*, 18 *Psychotria* cf. *alba*, 19 *Psychotria montivaga*, 20 *Palicourea luteonivea*

The general appearance of the forest becomes considerably wetter at higher altitudes due to lush epiphytic vegetation (i.e., increasing abundance of epiphytic bryophytes and vascular plants).

10.1.2.2.2 Forest Along Ridges and Upper Slopes from ca. 1900 m to 2100 m a.s.l. (Forest Type II, Subtype of B. Evergreen Lower Montane Forest)

This forest type attains 15–20 m on slopes but as little as 8–10 m along the most exposed ridges (Fig. 10.1.3). A characteristic feature of this forest type is the thick raw humus layer. Araceae are poorly represented; characteristic species are the bole climbing *Anthurium cutucuense* (shady upper slope forest) and the mostly terrestrial *A. grubbii* (open ridges). The herb and shrub layer is dominated by *Elaphoglossum* spp (Dryopteridaceae), *Cyathea bipinnatifida* (Cyatheaceae) and *Guzmania* spp (Bromeliaceae). Epiphytes are strikingly more abundant and speciose in the understorey than in forest type I. Epiphyte species characteristic for upper slope forest are *Hymenophyllum lobatoalatum* (Hymenophyllaceae), *Acronia microcardia*, *A. cf bivalvis*, *Pleurothallis lilijae* (Orchidaceae); characteristic in ridge forest are *Masdevallia gnomii* and congeners or *Specklinia* sp. nov. (Orchidaceae). Shared character species include *Mezobromelia capituligera*, *Guzmania killipiana* and *G. sanguinea* (Bromeliaceae) and *Brachycladium* spp (Orchidaceae).

In respect to tree species composition, forest type II is also quite different from the above-mentioned type I. Some frequent and characteristic species are *Alzatea verticillata* (Alzateaceae), *Dictyocaryum lamarckianum* and *Wettinia aequatorialis* (Arecaceae), *Weinmannia pinnata*, *W. sorbifolia* and *W. spruceana* (Cunoniaceae), *Abarema killipii* (Fabaceae), *Hyeronima moritziana* (Euphorbiaceae), *Ocotea aciphylla* (Lauraceae), *Graffenrieda emarginata* and *Miconia calophylla* (Melastomataceae), *Podocarpus oleifolius* (Podocarpaceae) and *Matayba inelegans* (Sapindaceae). While forest type I is strongly dominated by mesophyllous and straight trunk trees, ridge trees tend to be crooked and microphyllous (Paulsch 2002; see Chapter 10.3 in this volume). Among climbers the genus *Mikania* (Asteraceae) is well represented.

The bryophyte flora of this forest type is quite different from that of the previous one. *Plagiochila* and robust mosses characteristic of forest type I are less conspicuous, instead the epiphytic layers are dominated by members of *Bazzania* (e.g., *B. cuneistipula*, *B. hookeri*, *B. longistipula*, *B. phyllobola*, *B. stolonifera*), *Lepidozia cupressina* and the mosses *Leucobryum antillarum* and *Campylopus huallagensis*. Characteristic species of *Plagiochila* include the delicate *P. aerea* and *P. punctata* and the robust *P. tabinensis*. Ground-dwelling species are more abundant than in the previous forest type and include the mosses *Campylopus huallagensis*, *Leucobryum giganteum*, *Pyrrobryum spiniforme* and others.

10.1.2.2.3 Forest of Ridges and Upper Slopes at ca. 2100–2250 m a.s.l. (Forest Type III, Subtype of C. Evergreen Upper Montane Forest)

With increasing altitude, the canopy successively opens and trees do not surpass 12 m (ridge) and 15 m (slope) in height, respectively (Fig. 10.1.4). Epiphytes are abundant and highly diverse, last but not least close to the ground where they constitute

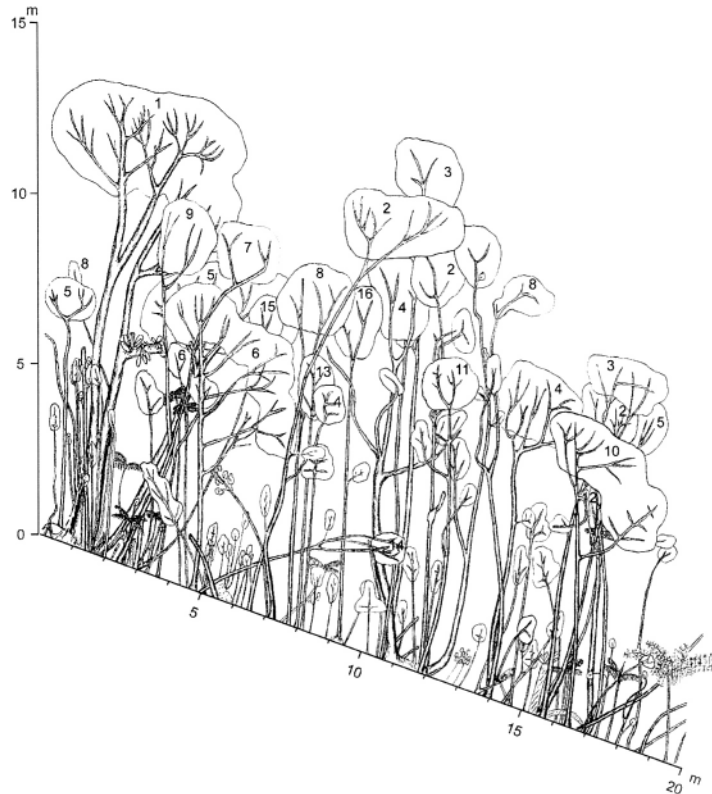


Fig. 10.1.3 Profile diagram of forest type II (Forest along ridges and upper slopes at ca. 1900–2100 m a.s.l., subtype of B. Evergreen lower montane forest) at the RBSF (2050 m a.s.l., depth of the profile is 5 m). Trees with diameters exceeding 5 cm: 1 *Alchornea grandiflora*, 2 *Hyeronima moritziana*, 3 *Weinmannia pinnata*, 4 *Clusia* sp., 5 *Matayba ineleans*, 6 *Graffenrieda emarginata*, 7 *Ocotea benthamiana*, 8 *Ocotea* sp., 9 *Persea* sp., 10 *Lauraceae*, 11 *Hedyosmum anisodorum*, 12 *Miconia* cf. *calophylla*, 13 *Podocarpus oleifolius*, 14 *Rudgea* sp., 15 *Sloanea* sp., 16 *Ilex* cf. *amboroica*

abundant of moisture-demanding (e.g., *Hymenophyllum ruizianum* (Hymenophyllaceae), *Semiramisia speciosa* (Ericaceae)), generalistic and (especially on ridges) heliophilous species (*Tillandsia laminata*, *T.* cf. *aequatorialis* (Bromeliaceae), *Melpomene moniliformis* (Grammitidaceae) and *Utricularia jamesonii* (Lentibulariaceae) along with numerous orchid taxa).

Common tree species are *Hedyosmum translucidum* (Chloranthaceae), *Clusia* cf. *ducuoides*, *Clusia* spp. and *Tovomita weddelliana* (Clusiaceae), *Weinmannia haenkeana* and *W. ovata* (Cunoniaceae), *Purdiaea nutans* (Clethraceae), *Alchornea grandiflora* (Euphorbiaceae), *Endlicheria oreocola*, *Licaria subsessilis*, *Ocotea benthamiana* and *Persea subcordata* (Lauraceae), *Eschweilera sessilis* (Lecythidaceae), *Graffenrieda emarginata* (Melastomataceae), *Calypttranthes pulchella* and *Myrcia* sp. (both Myrtaceae) and *Podocarpus oleifolius* (Podocarpaceae).



Fig. 10.1.4 Profile diagram of forest type III (Forest of ridges and upper slopes at ca. 2100–2250 m a.s.l., subtype of C. Evergreen upper montane forest) at the RBSF (2210 m a.s.l., depth of the profile is 5 m). Trees with diameters exceeding 5 cm: 1 *Graffenrieda emarginata*, 2 *Clusia* sp. 1, 3 *Podocarpus oleifolius*, 4 *Calyptanthes pulchella*, 5 *Alchornea grandiflora*, 6 *Myrcia* sp., 7 *Hyeronima moritziana*, 8 *Clusia* sp. 2, 9 *Dioicodendron dioicum*, 10 *Eschweilera sessilis*, 11 *Palicourea angustifolia*, 12 *Palicourea loxensis*, 13 *Faramea glandulosa*, 14 *Hedyosmum translucidum*, 15 *Purdiaea nutans*, 16 *Siphoneugena* sp., 17 *Matayba inelegrans*

The bryophyte flora of this forest type is rather similar to that of the previous one but is somewhat richer in heliophytic species such as the moss *Macromitrium trichophyllum* and the liverwort *Herbertus pensilis*, due to the more open and lower forest canopy (Parolly and Kürschner 2004b). The latter authors also recorded the moss *Leptotheca boliviana* and the liverwort *Bazzania aurescens* (earlier identified as *B. tricrenata*) as characteristic of this forest type.

10.1.2.2.4 Ridge Forest at ca. 2250–2700 m a.s.l., up to the Timberline (Forest type IV, Subtype of C. Evergreen Upper Montane Forest)

With increasing altitude the canopy further decreases in cover and height (5–10 m). Larger trees are typically scattered, closed canopy is restricted to saddles (Fig. 10.1.5). *Purdiaea nutans* (Clethraceae) monodominates this vegetation type up to ca. 2700 m. The forest ground is densely covered by terrestrial bromeliads, especially *Guzmania paniculata*.

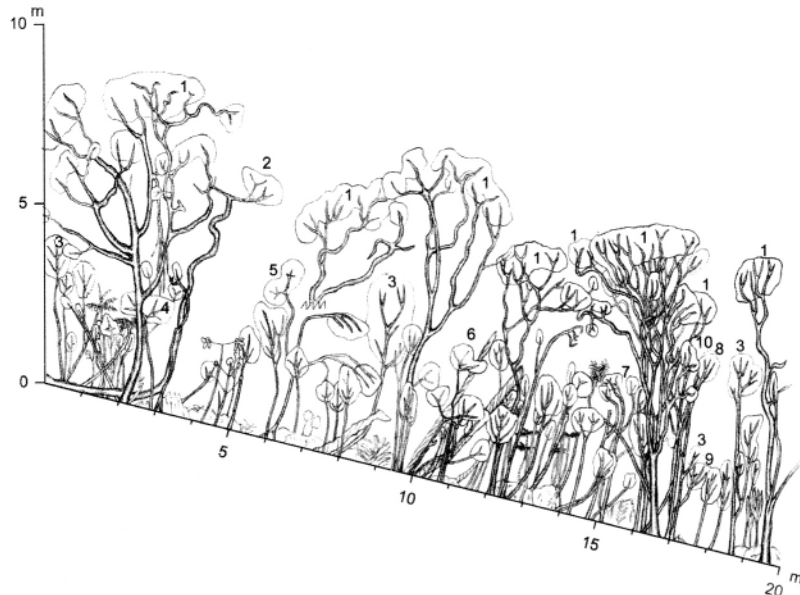


Fig. 10.1.5 Profile diagram of forest type IV (Ridge forest at ca. 2250–2700 m a.s.l., subtype of C. Evergreen upper montane forest) at the RBSF (2450 m a.s.l., depth of the profile is 5 m). Trees with diameters exceeding 5 cm: 1 *Purdiaea nutans*, 2 *Persea feruginea*, 3 *Graffenrieda harlingii*, 4 *Symplocos bogotensis*, 5 *Ternstroemia cleistogama*, 6 *Weinmannia fagaroides*, 7 *Freziera campanulata*, 8 *Clusia* sp., 9 *Weinmannia ovata*, 10 *Styrax trichostemon*

Obviously related to the high level of exposure to light, the understorey features a great abundance of canopy epiphytes such as *Racinaea dielsii*, *Vriesea fragrans*, *V. lutheri* (Bromeliaceae), *Maxillaria adendrobium*, *M. aggregata*, *Specklinia rubens*, *Trichosalpinx lenticularis* and *T. scabridula* (Orchidaceae). More hygrophilous plants such as *Racinaea seemannii* (Bromeliaceae), *Hymenophyllum ruizii* (Hymenophyllaceae) or *Lepanthes mucronata* (Orchidaceae) are restricted to particularly sheltered spots, like dense tangles.

Characteristic trees and shrubs are *Ilex rimbachii*, *I. scopulorum*, *I. teratopis* and *I. weberlingii* (Aquifoliaceae), *Weinmannia elliptica*, *W. fagaroides* and *W. loxensis* (Cunoniaceae), *Licaria subsessilis*, *Ocotea infravoveolata* and *Persea ferruginea* (Lauraceae), *Gaiadendron punctatum* (Loranthaceae), *Brachyotum campanulare*, *Graffenrieda harlingii* (Melastomataceae), *Cybianthus marginatus* (Myrsinaceae), *Calyptranthes pulchella* and *Myrcia* spp (Myrtaceae), *Podocarpus oleifolius* (Podocarpaceae), *Freziera karsteniana* (Theaceae) and *Drimys granadensis* (Winteraceae).

Undoubtedly, the most remarkable bryophyte feature of the *Purdiaea* forest is the abundance of the rare liverworts *Pleurozia paradoxa* and *P. heterophylla* (Pleuroziaceae; see Chapter 19 in this volume). *Pleurozia heterophylla* (only three

known localities worldwide) grows on trunks and branches, whereas *P. paradoxa* grows on tree bases; the latter extends upwards into páramo. Because of the low and open canopy, the epiphytic community of the *Purdiaea* forest is rich in Macromitriaceae (*Groutiella*, *Macromitrium*, *Schlotheimia*; ten species), which are typical canopy mosses of montane forest (Gradstein et al. 2001). Further characteristic epiphytes include members of the genus *Herbertus* (*H. divergens*, *H. juniperoides*, *H. sendtneri*), *Scapania portoricensis*, *Lepicolea pruinosa*, *Bazzania bidens*, *B. longistipula*, *Plachiochila bifaria*, *P. pachyloma*, *P. tabinensis* and the mosses *Holomitrium sinuosum* and *Bryohumbertia filifolia* (Parolly and Kürschner 2004b). Canopy twigs harbour numerous tiny liverworts of Lejeuneaceae. The terrestrial layer can be very conspicuous and is made up of species of *Bazzania* (e.g., *Bazzania gracilis*, *B. hookeri*, *Bazzania* sp. nov.), *Syzygiella* (e.g., *S. anomala*, *S. tonduziana*), *Lepidozia* (*L. caespitosa*, *L. macrocolea*, *L. squarrosa*) and in wet places *Sphagnum*.

10.1.2.2.5 Subpáramo (“Ceja de Montaña”) Above the Timberline at ca. 2700 m (3000 m) to 3150 m a.s.l. (Subtype of D. Evergreen Elfin Forest)

The vegetation above the timberline is largely dominated by shrubs (Fig. 10.1.6). Some characteristic species are *Loricaria* sp. (Asteraceae), *Clethra ovalifolia* (Clethraceae), *Weinmannia cochensis* (Cunoniaceae), *Desfontainia spinosa* (Loganiaceae), *Brachyotum* spp, *Miconia* spp (Melastomataceae), *Myrica pubescens* (Myricaceae), *Styrax foveolaria* (Styracaceae) and *Symplocos sulcinervia* (Symplocaceae).

Terrestrial herbs become more speciose at these altitudes; some common species are *Dorobaea* sp. (Asteraceae), *Puya eryngioides*, *P. roseana* and *Guzmania lychnis* (Bromeliaceae), *Paepalanthus meridensis* (Eriocaulaceae), *Calamagrostis macrophylla*, *Chusquea falcata* and *Neurolepis elata* (all Poaceae) and *Arcytophyllum* spp (Rubiaceae).

The bryophyte flora of the subpáramo is very distinct although still poorly studied. Characteristic trunk-inhabiting species include the endemic moss *Macromitrium perreflexum* (abundant!) and the liverworts *Herbertus acanthelius*, *H. pensilis*, *Jamesoniella rubricaulis*, *Lepicolea pruinosa*, *Leptoscyphus gibbosus* and *Pleurozia paradoxa*. Several of these may also occur on soil. Twigs are inhabited by a unique community of tiny liverwort species including four that were described as new to science from the RBSF: *Physotheca autoica* (new monotypic genus), *Plagiochila cucullifolia* var. *anomala*, *Diplasiolejeunea eros-trata* and *D. grandirostrata*. The terrestrial layer is well developed and made up of species of *Riccardia* (notably *R. fucoidea*), *Isotachis* (*I. lopezii*, *I. muticeps*), *Sphagnum* (species still unidentified), *Nowellia curvifolia*, the robust mosses *Polytrichadelphus aristatus* and *Rhacocarpus purpurascens* and some of the trunk epiphytes mentioned above.

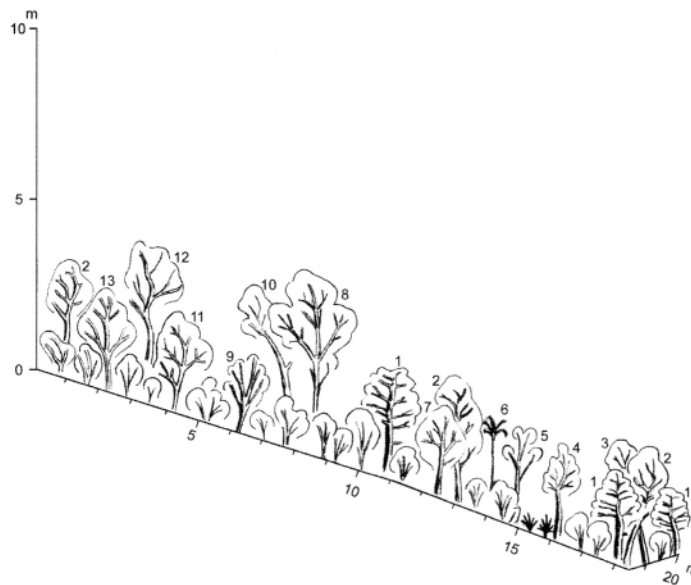


Fig. 10.1.6 Profile diagram of the subpáramo [Subpáramo (ceja de montaña) above the timberline at ca. 2700m (3000m) to 3150m a.s.l., subtype of D. Evergreen elfin-forest] at the RBSF (2750m a.s.l.). Trees with diameters exceeding 5 cm: 1 *Podocarpus oleifolius*, 2 *Clusia cf. duc-uoides*, 3 *Weinmannia loxensis*, 4 *Calypttranthes cf. pulchella*, 5 *Schefflera* sp., 6 *Geonoma* sp., 7 *Myrica pubescens*, 8 *Meriania rigida*, 9 *Macleania* sp., 10 *Weinmannia fagaroides*, 11 *Miconia cf. jahnii*, 12 *Weinmannia elliptica*, 13 *Graffenrieda harlingii*

10.1.2.2.6 The Forests of the RBSF in the Neotropical Context

Compared with many other neotropical montane forests the RBSF is peculiar in several aspects. The canopy along upper slopes and ridges is remarkably low and open across the entire elevational gradient. This points to a limitation of plant growth as a result of unfavourable environmental conditions.

The average canopy height of 6–8m found on ridges at 2300–2500m a.s.l. (Homeier 2004) is very low compared with other montane forests in Ecuador even at considerably higher elevations (e.g., ridge forest at Yanayacu Biological Station in North Ecuador attains 15m at 2900m a.s.l.; J. Homeier and F.A. Werner, personal observation). The timberline is very low (2700–3000m a.s.l.) and the genus *Polylepis* (Rosaceae), usually constituting the timberline of other Andean forests, is missing in South Ecuador (see Chapter 10.4 in this volume).

Comparable neotropical mountain areas with similar stunted, microphyllous forests include the Cordillera del Condor (located just east of Podocarpus NP), the Guyana shield (including the Venezuelan tepuis), and northern Peru (Alto Mayo). Apart from structural features these areas share certain plant taxa mainly restricted to sandstone. The most notable example is the tree *Purdiaea nutans* (Clethraceae),

which dominates the upper slopes and ridges of the RBSF forests above 2100 m a.s.l. (see Chapter 19 in this volume). Other species are *Quiina yatuensis* (Quiinaceae), a tree species until recently only known from Venezuelan Guyana, or the fern *Pterozonium brevifrons* (Pteridaceae).

At the family level, however, floristic composition is quite similar to other Neotropical montane forests, with Lauraceae, Melastomataceae and Rubiaceae being the most speciose tree families as, e.g., reported by Webster and Rhode (2001) for Maquipucuna reserve, North-west Ecuador (1100–2800 m a.s.l.) or by Dorr et al. (2000) from Guaramacal, Venezuela (1500–2800 m a.s.l.).

The plant species richness at RBSF is high. Several taxa show surprisingly high species numbers compared with other neotropical montane forests, although comparable floristical data from neotropical montane forests are still scarce.

Exceptionally high local species richness has been confirmed for some taxa such as the orchids in general (C.H. Dodson, personal communication) and the genus *Stelis* in particular with ca. 40 species (C. Luer, personal communication), Lauraceae with more than 40 species (H. van der Werff, personal communication) and *Piper* with over 30 species.

Small-scale environmental heterogeneity boosts the development of floristically and structurally distinct vegetation types and thus contributes to the outstanding plant species richness of the area as well as the region as a whole. In addition to the processes typical for primary forests, such as gap dynamics (see Chapter 23 in this volume) and the common landslides (see Chapter 24), steep climatic gradients (see Chapter 8) allow a wide array of trees to coexist and form a mosaic of physiognomic distinct vegetation units. These, in turn, amplify the abiotic heterogeneity to create an even wider magnitude of microhabitats and regeneration niches for other life forms. The combination of these processes together with the high topographic diversity and the related edaphic patchiness (see Chapter 9) maintain a high environmental heterogeneity throughout space and time.

10.1.3 Conclusions

In conclusion, the extraordinarily broad matrix of environmental conditions in the RBSF and its wider surroundings fosters high beta diversity allowing the coexistence of habitat specialists at the smallest spatial scale. The geographical position of the reserve in the ecotone between the hyperhumid eastern Andean slope and the dry interandean forests of South Ecuador on the one hand and the central and the northern Andes on the other hand provides access to an immense species pool communicated by the unique regional depression of the Andes.